Dylan Bruce

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Education

Georgia Institute of Technology **Expected Fall 2026**
B.S. in Computer Science, Minor in Mathematics *Atlanta, GA*

- **GPA**: 3.91/4.00
- **Activities**: Big Data Big Impact @ GT, WebDev @ GT, Volunteering, Spikeball

Experience

Machine Learning Researcher Aug. 2025 – Present
Georgia Tech VIP Aquabots: Marine Robotics *Atlanta, GA*

- Increased InceptionV4 model accuracy by approximately 10% through enhanced preprocessing and data normalization for plankton image classification
- Collaborating with a multidisciplinary team to develop and evaluate advanced preprocessing pipelines for underwater imagery

Teaching Assistant - Linear Algebra / Discrete Mathematics May 2025 – Present
Georgia Tech Department of Mathematics *Atlanta, GA*

- Led instruction and review for 1,000+ students in Linear Algebra and Discrete Mathematics, covering proofs, vector spaces, matrices, and algorithmic complexity
- Mentored students in rigorous problem-solving, boosting exam performance by ∼10%

Projects

FishCast - Climate-Fishery Forecasting Sep. 2025 - Present

- Analyzing climate and fishery datasets using RNNs (LSTMs) in PyTorch and Python to forecast shifts in marine species distributions
- Collaborating with a cross-functional team to develop ML models with Georgia Tech PACE Supercomputing

News2Lang - Context-Aware Language Learning Platform Sep. 2025 - Present

- Developing an AI-powered language learning platform with WebDev@GT to use NLP and LLMs to extract phrases, highlight context, and provide simplified explanations
- Building application with Next.js + Tailwind frontend and Supabase backend
- Integrating OpenAI API for context-aware explanations

Local Rental Application Aug. 2025 - Present

- Developing a full-stack React.js application for local property rental management, featuring real-time chat and booking
- Implementing backend services with Firebase for authentication, database, and hosting
- Designing responsive UI components using Tailwind CSS, improving usability across devices

Movie Success Prediction Model Feb. 2025 - Apr. 2025

- Built linear regression and neural network models in Python to predict box office revenue and wrote tests to evaluate an average ∼14% mean absolute error in accuracy
- Used scikit-learn and Keras to model and feature engineered from 5,000+ movie metadata entries

Technical Skills

Languages: Python, SQL, C, C#, Java, JavaScript, HTML/CSS, C++
Tools/Frameworks: Git, Docker, Firebase, React.js, Next.js, Tailwind.css, Android Studio, LaTeX
Machine Learning: TensorFlow, XGBoost, scikit-learn, Keras, Pandas, Matplotlib, Seaborn, NumPy, Excel
Technical: Data Structures, Machine Learning, Object-Oriented Programming, Optimization, A/B Testing, Agile/Scrum